

Christos Hadjinikolis

Engineering Manager | ML Systems Lead

Ph.D. Computer Science | Production ML, AI Standards & GenAI Harness Work

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about me

profiles

website

LinkedIn

GitHub

Google Scholar

leadership

6 direct reports

10-person team

hiring/coaching

delivery ownership

AI/product

production ML

model-serving APIs

real-time inference

evaluation/replay

GenAI

LLM/tool use

agent harness

context boundaries

approvals/traces

quality

observability

debugging

runbooks/rollback

shared on-call

engineering

Python/Java

Kotlin/SQL

Kafka/Flink

MLflow/Airflow

CI/CD

open source

PyPI: dynamicio

skeleton-replay

JetBrains plugin

I am an **Engineering Manager and ML systems lead** with 16 years across AI research, Data Science, ML Engineering, backend systems, and production platform leadership. I manage teams while staying technically involved in architecture, delivery, and operational quality. At Vortexa I lead a **10-person** cross-functional team and own a live ML/streaming estate that turns **6M** vessel-position records/hour into production intelligence for **13.5K** monitored vessels. My deepest production experience is in reliable ML systems; recent private GenAI work gives hands-on familiarity with agentic workflows, tool use, context boundaries, approvals, traces, and replay.

work-experience

Vortexa Ltd, London, UK 12/2020–present

Engineering Manager | ML Systems Lead

Own engineering strategy and delivery for Vortexa's live post-ingestion ML/data estate: model-serving APIs, real-time event pipelines, evaluation loops, monitoring, and downstream product interfaces. Lead a **10-person** cross-functional team across engineering, Product, and domain expertise, with **6** direct reports and accountability for model quality, reliability, roadmap execution, and delivery maturity.

- **People & Delivery:** Manage **6** direct reports spanning Product ownership, SME analysis, Data Science, and Data Engineering; own performance/career development, team delivery, hiring for **6** roles, and delivery accountability for the wider team.
- **Roadmap & Product Alignment:** Partner with Product, analysts, SMEs, and engineers to turn ambiguous prediction-quality and product-semantics problems into measurable objectives, semantic contracts, interface/SLA proposals, and OKR-linked workstreams.
- **Production ML Systems:** Led research-to-production delivery for destination and ETA/waiting-time prediction services; established **MLflow**, model/data versioning, service deployment, replay/evaluation interfaces, live accuracy monitoring, and model-serving workflows.
- **Evaluation & Answer-Quality Analogue:** Built batch/online model-quality and replay loops; analysed edge cases and failure modes with domain experts and Product to drive metric-guided model, data, and product-interface improvements.
- **Architecture & Scale:** Led **Kafka Streams-to-Flink** as a strategic move for critical live pipelines, improving operational visibility, state/checkpointing, maintainability, and compute scaling independent of Kafka partitioning.
- **Testing, Observability & Reliability:** Raised engineering quality through local E2E/integration tests, ADRs, dev containers, runbooks, rollback/fallback paths, shared on-call, and monitoring; maintained **MTTR** under **30** min.
- **Developer Clarity:** Standardised I/O seams, data contracts, repo archetypes, AWS CodeArtifact publishing, and reusable pipeline patterns; cut a **3h** DAG feedback loop to under **5** min.

DataReply, London, UK 04/2016–12/2020

Senior Consultant | Data Scientist / ML Engineer

Joined Data Reply's London spin-off as its first consultant and helped grow it from **three practitioners and two managers to 30+ consultants**. Progressed to Senior Consultant; shaped opportunities, agreed deliverables/timelines, managed expectations, placed consultants, and delivered technical roadmaps across UBS, Vodafone, CNHi, and other enterprise clients.

- **Client Delivery Leadership:** Led client meetings, scoped goals, agreed deliverables, managed expectations, supported hiring/consultant placement, and mentored colleagues across banking, telecoms, manufacturing, publishing, and gaming accounts.
- **Vodafone Group:** Led multi-workstream GCP data-science platform work based on Kubeflow; shaped requirements, migration planning, CI/CD, workflow automation, ML lifecycle support, and feature-engineering migration.
- **CNHi:** Lead Data Scientist/Scrum Master on predictive maintenance with Azure/Databricks; translated stakeholder needs into a DS/DE backlog and guided a **PySpark** team over live telemetry.
- **UBS:** Embedded consultant who grew from Data Scientist into ML Engineer through XP/pairing while delivering graph/process-mining analytics, observability, and real-time **Kafka**/Elasticsearch pipelines.

KCL, UCL, David Game College, UK 2010–2016

Teaching and Technical Training

Delivered courses and tutorials in algorithm design, data structures, databases/SQL, and Java/Python during and after doctoral research; developed presentation discipline, learner support, and mentoring habits.

education

2010–2014	Ph.D. in Computer Science <i>Thesis: Persuasion Dialogues & Opponent Modelling.</i> Developed logical inference methods for large Knowledge Graphs , combining symbolic reasoning, graph analytics , and Bayesian techniques (e.g., MCMC). Received Best Poster Award at IJCAI 2013 (413 submissions), EPSRC PhD Scholarship , Outstanding TA Award , and Graduate Certificate in Academic Practice . Access Publications on Google Scholar	King's College London, UK
2004–2010	Diploma (BEng) in Computer Engineering Five-year polytechnic degree in Computer Science and Engineering with a strong focus on mathematics (10 courses), covering theoretical aspects, networking, and telecommunications. Majored in Artificial Intelligence .	University of Thessaly, GR
2002–2004	Military Officer: Second Lieutenant Completed national service as a commissioned Second Lieutenant.	National Guard, CY

selected AI/tooling work

Private	Promet Private local-first AI harness work around memory-aware, voice-enabled, tool-using assistants: local runtimes, faster-whisper STT, Kokoro/ Hugging Face assets, streaming interaction boundaries, tool routing, schema validation, approval gates, durable state, traces, and replay/evaluation for auditable LLM interactions.	Local GenAI harness
PyPI	skeleton-replay Python developer tool that turns script/pytest runs into traces, architecture snapshots, workflow evidence, and replayable reports for review, debugging, onboarding, and LLM-assisted code understanding.	Runtime evidence
JetBrains	Skeleton Replay plugin Free PyCharm/IntelliJ plugin bringing Skeleton reports, workflow evidence, and source navigation into the IDE; reflects a bias for developer experience, evidence, and shared understanding.	IDE workflow
PyPI	dynamicio Free public tooling for making I/O seams explicit, validating schema/data expectations, and switching local/dev/prod datasets so pipeline logic can be tested with faster feedback.	Data contracts

committees/affiliations

since 10/2024	University College London, Dept. of Information Studies Research/teach practical AI standardisation: AI Act, auditability, model/data versioning, explainability, and safe adoption.	Associate Researcher
since 01/2021	ISO - International Organisation for Standardisation Member of JTC'21 Working Group 3, developing AI standards aligned with EU policy and international norms.	Committee Expert Member for JTC'21

talks/interviews

2023	Bill Raymonds, Agile in Action Agile Data Science: Sailing Through Data Science: The Agile Journey at Vortexa, Details: YouTube	Podcast Interview
2022	Open Data Science Conference dynamicio , a published PyPI library for abstracting I/O in ML/data workflows, Details: here	Industry Conference Talk
2020	iunera: Big Data Innovators The Agile approach in data science explained by an ML expert, Details: here	Interview Blog Post
2020	Big Data Warsaw Monitoring & Analysing Communication and Trade Events as Graphs, Details: here	Industry Conference Talk
2018	Connected Data London AI tribes and interconnections: what's graph got to do with it?, Details: here	Panelist @ Industry Conference
2018	Minds Mastering Machines A talk on doing Data Science the Agile Way, Details: here	Industry Conference Talk

certificates & awards

2017–2020	Selected ML/cloud credentials AWS ML Specialty; Google Professional Data Engineer; Apache Spark Developer.	AWS, Google Cloud, O'Reilly
2016–2019	Data/graph engineering credentials Process Mining; Graph Analytics; Advanced Elasticsearch; Neo4j Certified Professional.	Coursera, Elastic, Neo4j